

VEDAANG CHOPRA

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MS CS (ML) at Georgia Tech — Advised by **Dr. Matthew Gombolay**, CORE Robotics Lab

Research: Proactive multimodal agentic systems – Computer use agents with video/audio/image/text/keypoint grounding – VLM routing & evaluation

EDUCATION

Georgia Institute of Technology

Atlanta, GA

M.S. in Computer Science (Machine Learning Specialization), GPA: 4.0/4.0

Aug 2024 – Dec 2026

– **Coursework:** Deep Learning, Deep RL, ML Security, Large & Vision Language Models, Agentic AI, Systems for AI

Maharaja Surajmal Institute of Technology

New Delhi, India

B.Tech. in Information Technology, CGPA: 8.8/10.0

Aug 2016 – Aug 2020

RESEARCH EXPERIENCE

Georgia Institute of Technology – CORE Robotics Lab @ Siemens

Atlanta, GA

Graduate Research Assistant — Advisor: Dr. Matthew Gombolay

Jan 2026 – Present

- Building a **proactive multimodal agentic pipeline** for parametric CAD code generation: LLM generates CadQuery programs, execution environment returns compile/geometry feedback, LangGraph-orchestrated repair loop iteratively corrects errors – framing synthesis as inference-time search with verifiable geometric reward.
- Designing **geometric verification** scoring compile validity, parametric 3D correctness (Chamfer/Hausdorff on STL/B-Rep), and repair trajectory quality; 202-test evaluation harness with **VLM-based visual judgment** and RAG-augmented prompts across 16-module system.

Georgia Institute of Technology – CS 8903 Special Problems

Atlanta, GA

Graduate Researcher

Jan 2025 – Present

- **ARTEMIS: Cost-aware VLM routing** dynamically selecting optimal VLM per request across VQA, OCR, captioning, reasoning; trained neural multi-task router with SLA-aware load balancing, serving 6+ VLM backends via vLLM – **multimodal routing at production scale**.
- **ATHENA: Multi-agent supervisor-worker orchestration** via LangGraph Command routing; 5 specialized agents with reflection-based critique and dynamic plan modification; evaluated with BLEU-4 (+0.18 absolute) and CLIPScore.
- **SHASTRA: Agent workflow retrieval framework** enabling compositional reuse of tool-call sequences for long-horizon tasks under cost/latency constraints; LangGraph planner with Pydantic state and pluggable component registry.

INDUSTRY EXPERIENCE

Fortinet Technologies Inc. (AIOps R&D)

Bengaluru, India

Software Development Engineer I & II (ML/AI)

Feb 2021 – Jul 2025

- **Led agentic RAG diagnostics system** (hackathon → production): LLM plans multi-step tool calls over network telemetry, invokes APIs, verifies hypotheses via structured function calling – autonomous root-cause analysis reducing mean resolution time 70%. *Shipped frontier agentic product to production.*
- Scaled OpenSearch ingestion 50 to 2,000 events/sec using Golang; integrated ML models into backend services with Python/Go across distributed telemetry infrastructure.
- Patent (Filed): PCT/IN2022/058026 – Unsupervised distributional thresholding for wireless connectivity anomaly detection; reduced manual troubleshooting 75%.

PUBLICATIONS & PATENTS

Patent (Filed): PCT/IN2022/058026 – Unsupervised distributional thresholding for wireless connectivity anomaly detection.

TECHNICAL SKILLS

Multimodal Agentic AI: Computer Use Agents, Proactive Multimodal Systems (Video/Audio/Image/Text/Keypoints), VLM Routing & Evaluation, Agentic RAG, Tool/Function Calling, Multi-Agent Orchestration, LangGraph, LangChain, Supervisor-Worker Patterns, Planning & Reasoning, Structured Generation (Pydantic)

LLMs / VLMs: vLLM, Vision-Language Models (CLIP, SBERT, FAISS), Cross-Modal Retrieval, Inference Optimization, Model Evaluation & Benchmarking, Post-Training Workflows

AI / ML: PyTorch, Hugging Face Transformers, Deep Learning, CNNs, Vision Transformers, NLP, Computer Vision, Scikit-Learn, NumPy, Pandas

Reinforcement Learning: PPO, DQN, Reward Shaping, Curriculum Learning, Self-Play (academic projects)

Systems: Python, Go, C/C++, SQL/PostgreSQL, Redis, OpenSearch/Elasticsearch, Docker, Linux, Git, ONNX Runtime, FastAPI, Azure, W&B, TensorBoard